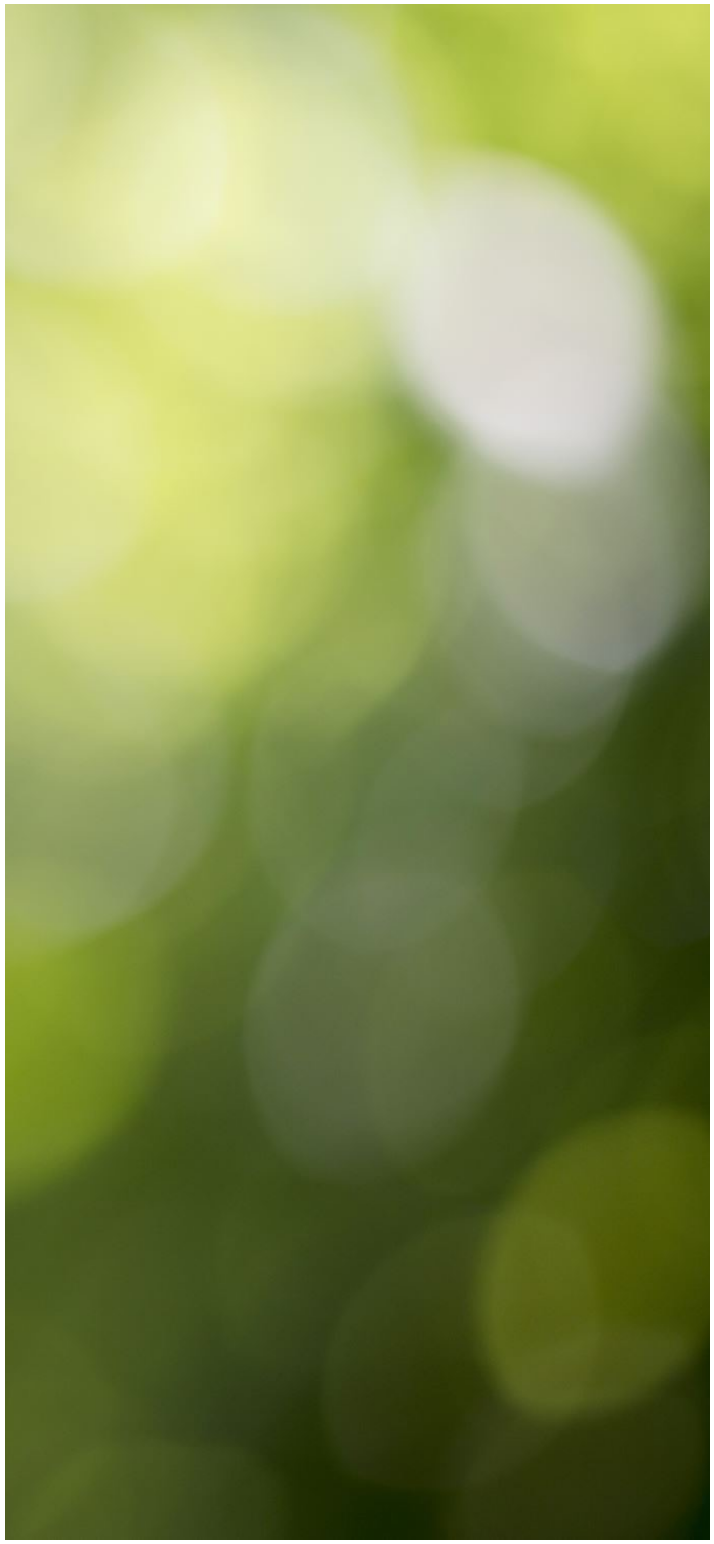
Mechatronics Design Laboratory I

ENME 561/661

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Associate Professor
Department of Mechanical and Manufacturing Engineering

2026/09/04



Who's in the Class?

Graduate Students

Undergraduate Students

- Mechanical,
- Electrical,
- Software and

Use of AI tools

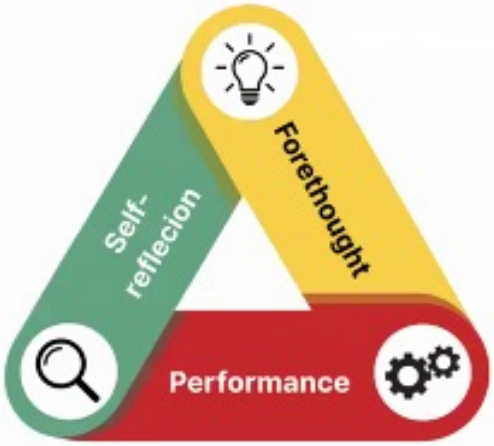
- Transparent Use: AI tools may be used to **support your learning** and content creation, with appropriate attribution.
- You may use AI for drafting, problem-solving support, or code assistance, provided that you **critically evaluate** and revise all outputs.
- **You are responsible** for the accuracy and integrity of your submission.
- All use of AI, including prompts and tool identification, must be clearly documented through an AI use statement included with your work.
- You may use GenAI to brainstorm topics or clarify programming concepts, but the analysis must be your own because I'm assessing your ability to learning course objectives.
- Using AI to generate your analysis or programming codes without citation misrepresents your work and prevents you from developing critical thinking skills.

Supporting Your Learning with AI: A Self-Regulation Approach

- It should support your learning and help you achieve the course outcomes
- Zimmerman Self-regulated Learning Model (explains how learners actively manage their own learning process):

After Learning

1. Self-judgement – I want to assess how much I've improved in academic writing since the start of term. Help me identify three ways my latest paper is stronger than my first.
2. Self-reaction – I'm feeling burned out. I still want to succeed in this course. Help me reduce my goals for the week without falling too far behind.



Before Learning

1. Task-analysis – Create a project timeline with milestones.
2. Motivational beliefs – I got a low mark on my first paper. Suggest three ways this can actually help me do better in the course.

During Learning

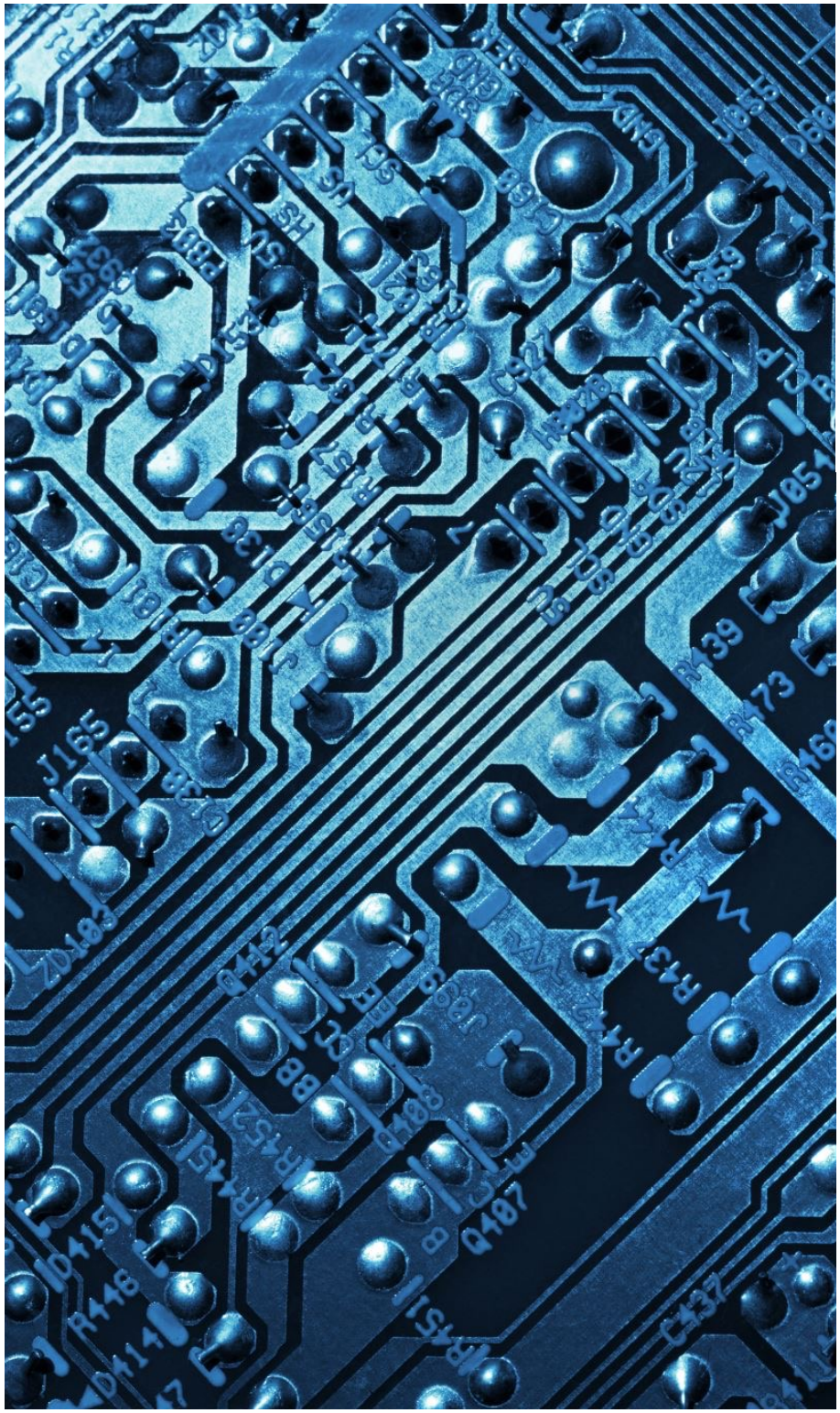
1. Self-control – *I just lost 45 minutes scrolling during my study time. Help me re-focus and make the most of the next hour.*
2. Self-observation – here are my *notes on three chapters of the child development textbook. Quiz me on my notes by asking multiple choice questions.*

Source: Dr. Soroush Sabbaghan's presentation & Becoming a Self-Regulated Learner: An Overview by Barry J. Zimmerman, 2016

UCalgary - Use of AI Resources

AI for students:
https://libguides.ucalgary.ca/c.php?g=733971&p=5278501

Citing AI tools:
https://libguides.ucalgary.ca/c.php?g=733971&p=5302331



Core Elements of Mechatronics

- Mechatronics:
  - Use of electronics in mechanical control
  - Application of complex decision-making to the operation of a physical system

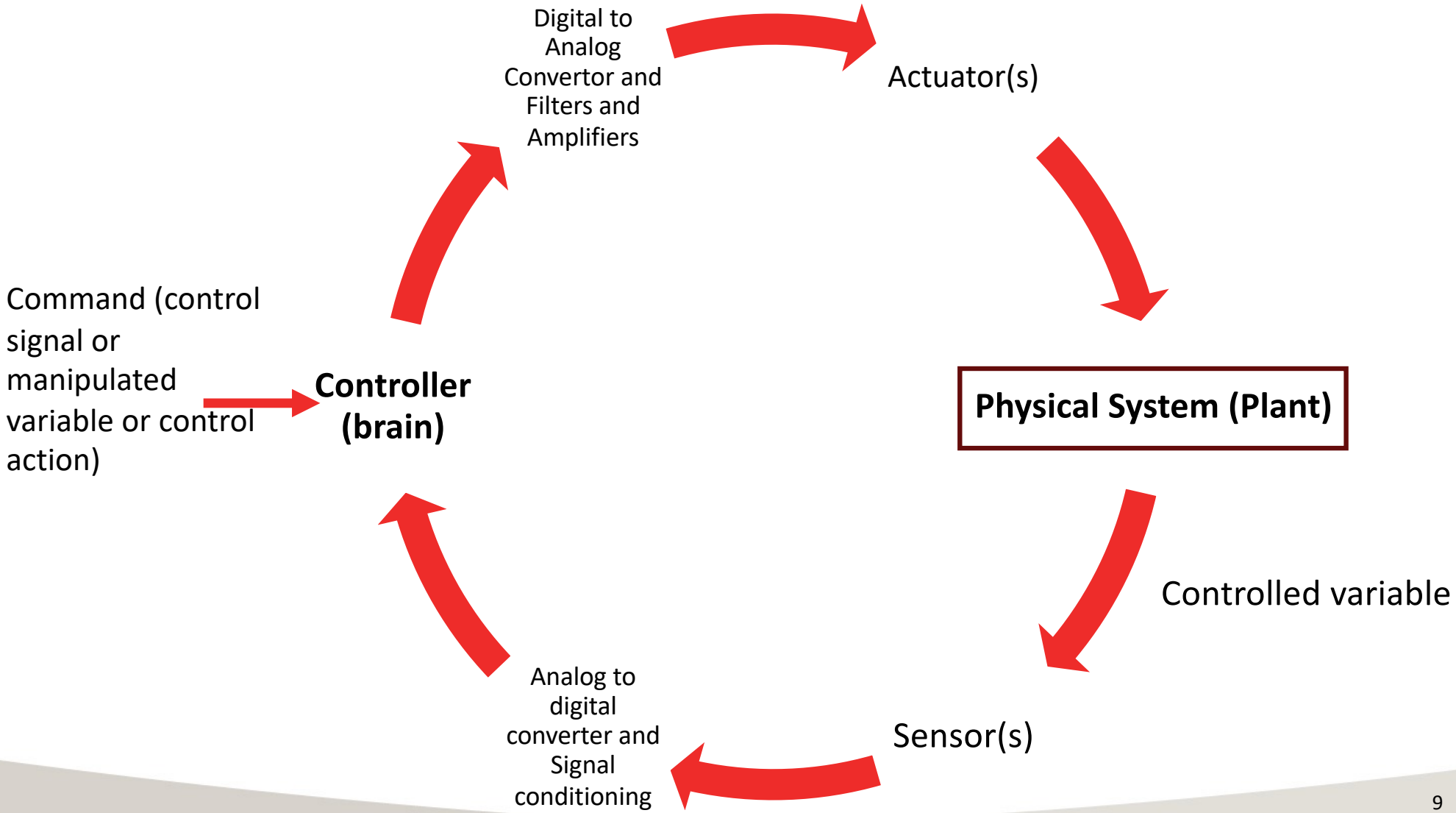
In this course, we will learn about

- Hardware
  - Sensors, actuators,…
- Software
  - To interact with hardware elements
- How to control mechatronics systems

Mechatronics: Application of complex decision making to the operation of physical systems

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Components of a Mechatronics System:



Lab Schedule

Day and Time	Section
MF 2:00AM - 3:50AM	561 B02
MF 4:00PM - 5:50PM	561 B03
TuTh 4:00PM - 5:50PM	561 B04
TuTh 2:00PM - 3:50PM	661 B01

4 lab sections, each run by a TA.

Each section = 4 hours/week → **16 hours total**.

Although I cannot attend all labs, I will:

✓ **Join some sessions** on a rotating basis

✓ **Be available** during lecture & office hours

✓ **Train & support TAs** regularly

✓ **Review lab progress & provide guidance** regularly

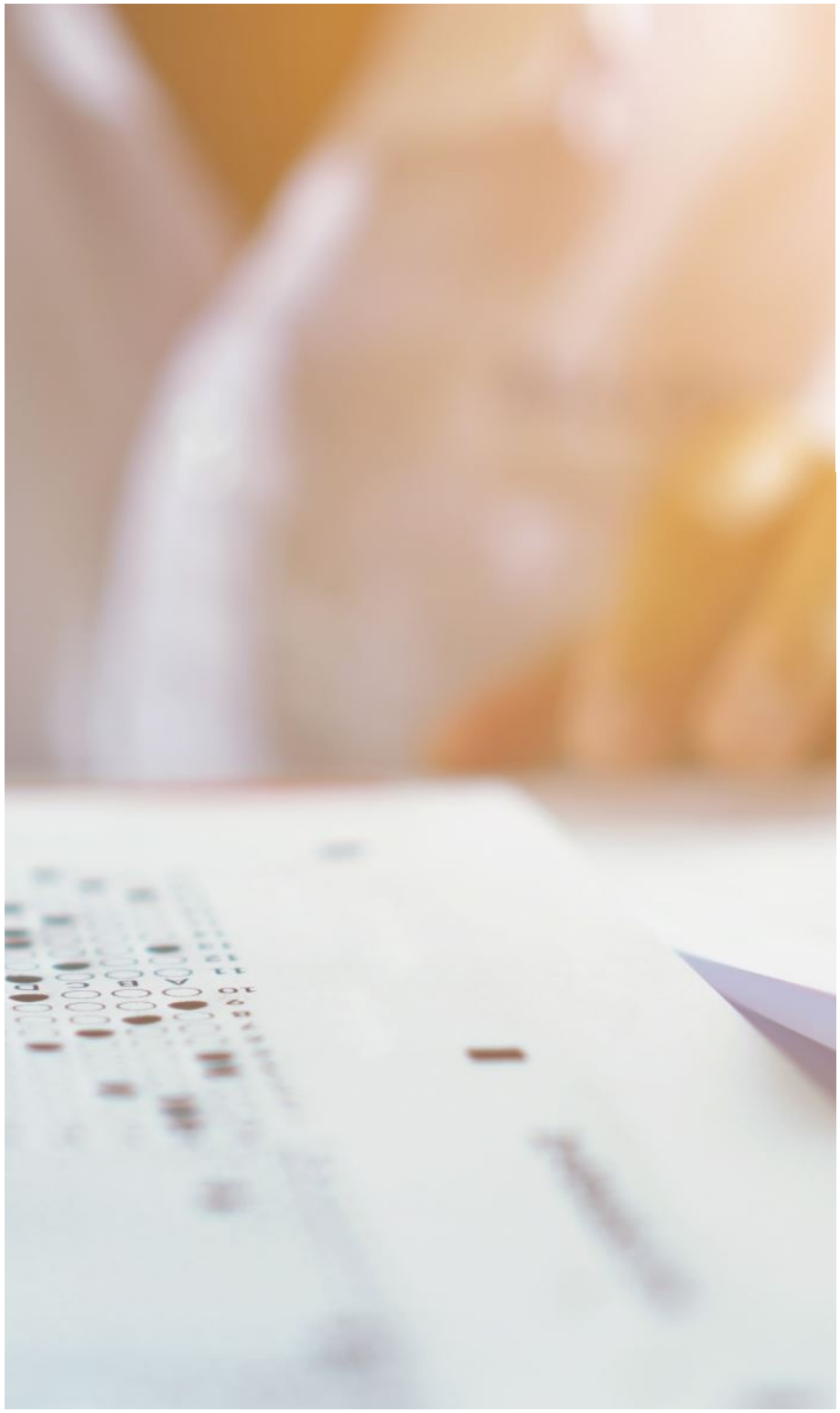
Lab	Topics	Submission format and deadline
1	MATLAB and Simulink	Solo (Sep 17)
2	Modelling: Block diagram, Transfer Function, Differential Equations	Group (Oct 1)
3	Friction Identification, Parameter Estimation	Group (Oct 15)
4	Practical Design of PID controllers	Group (Oct 29)
5	Digital Signal Acquisition	Group (Nov 19)
6	Passive and Active Filters	Group (Dec 3)

There are four sessions for each lab.
Lab handouts will be posted on Fridays.

Lab Attendance Policy

- Lab attendance is mandatory unless you have completed all the experiments and informed the TA.
- You must attend the lab sessions in which you are enrolled.
- If you need extra time, you may attend sessions with fewer than 8 groups. These extra sessions are on a first-come, first-served basis, and the TA's priority will be assisting students who are enrolled in that session.

Individual Exit survey after each lab



Assessment

- 6 Laboratory Reports (60%)
- Final Exam (40%) – In person
  - can bring 1-page notes into the exam room
- 5 Lab reports reflect group work, while the final exam assesses individual performance.
- If a student scores below 40% on the final exam, the maximum final grade they can receive is a “D.”

Lab 1

- Programming Questions
- Resources:
  - MATLAB Onramp Training
    - Functions, arrays, plots, …
    - Small project
  - Simulink Onramp Training
    - Blocks, Visualization of signals, Discrete systems, Continuous systems,..
    - Small projects
  - Simulink Training
  - Office Hours!

MATLAB and Simulink Installation on Your Personal Computer

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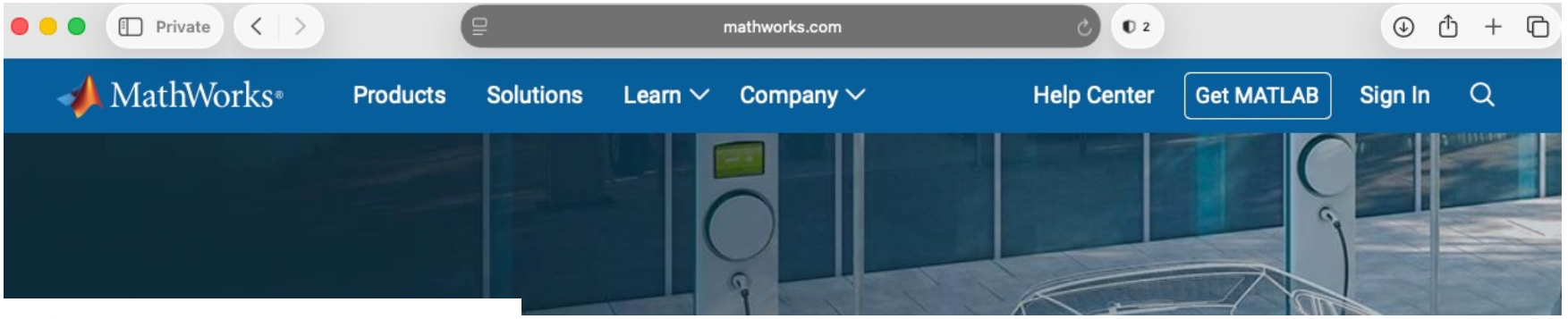
To run MATLAB outside of EEEL 103:

- Install MATLAB and Simulink on your personal computers
  - Free installation through the university can be found at:
    - https://iac01.ucalgary.ca/SDSWeb/Public/SoftwareList.aspx
    - Need Control Systems Toolbox and Simulink
      - Can be installed through the Add-Ons window if not already installed
- To work with hardware in EEEL 103, you will be provided with laptops with pre-installed, specialized software that cannot be installed on personal computers.

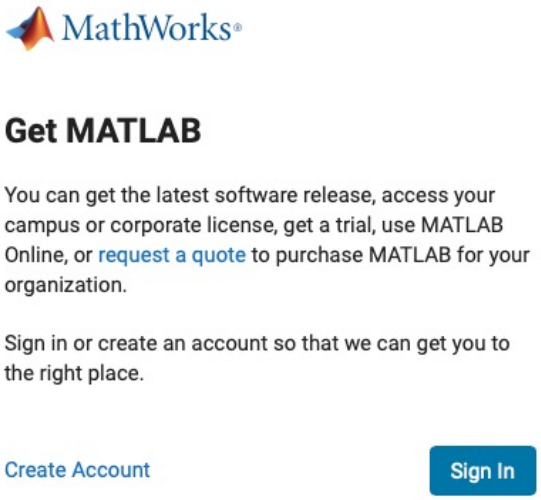
If you do not have access to a computer to run MATLAB, please email me

MATLAB and Simulink Installation

1



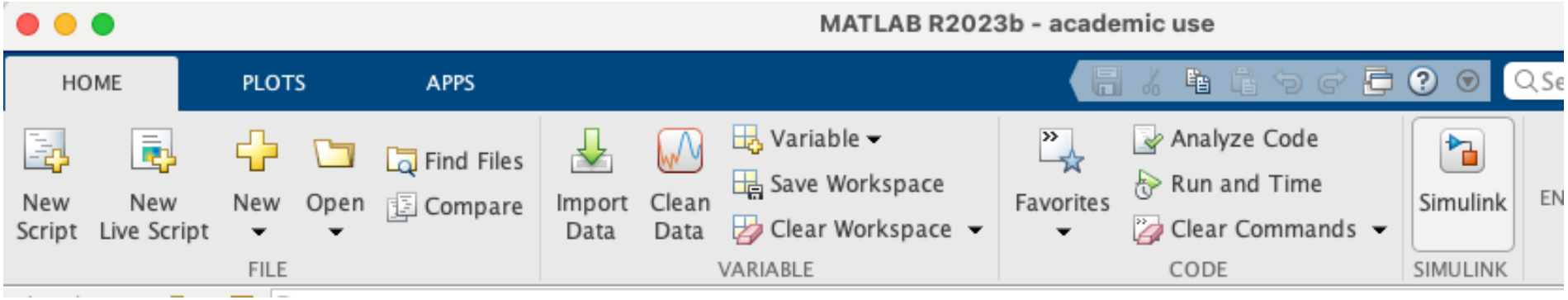
2



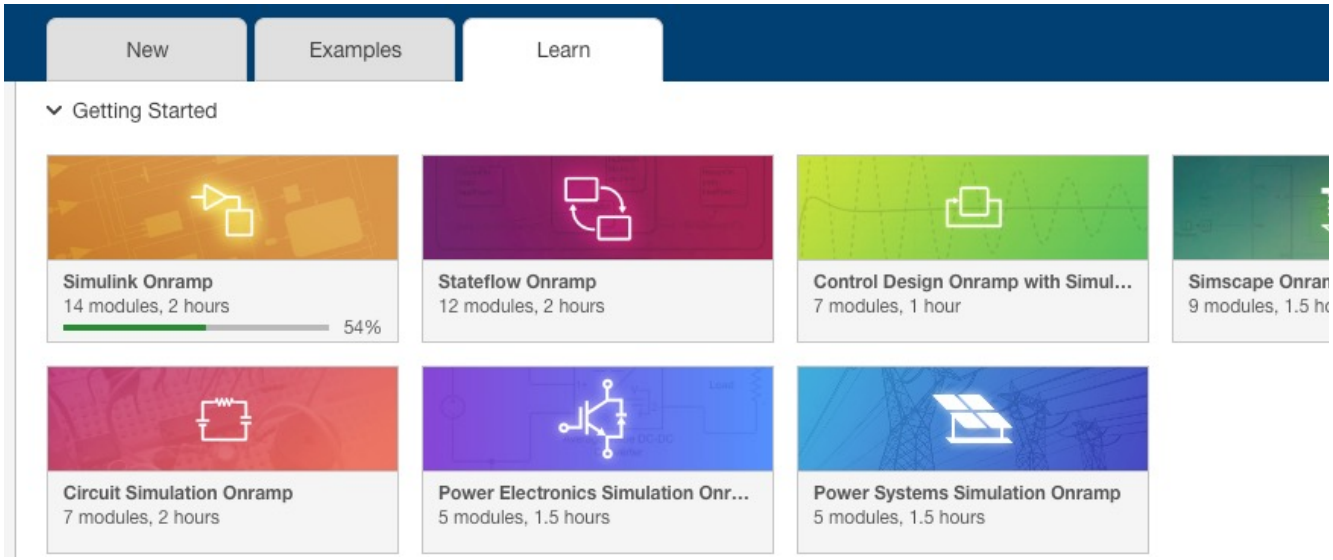
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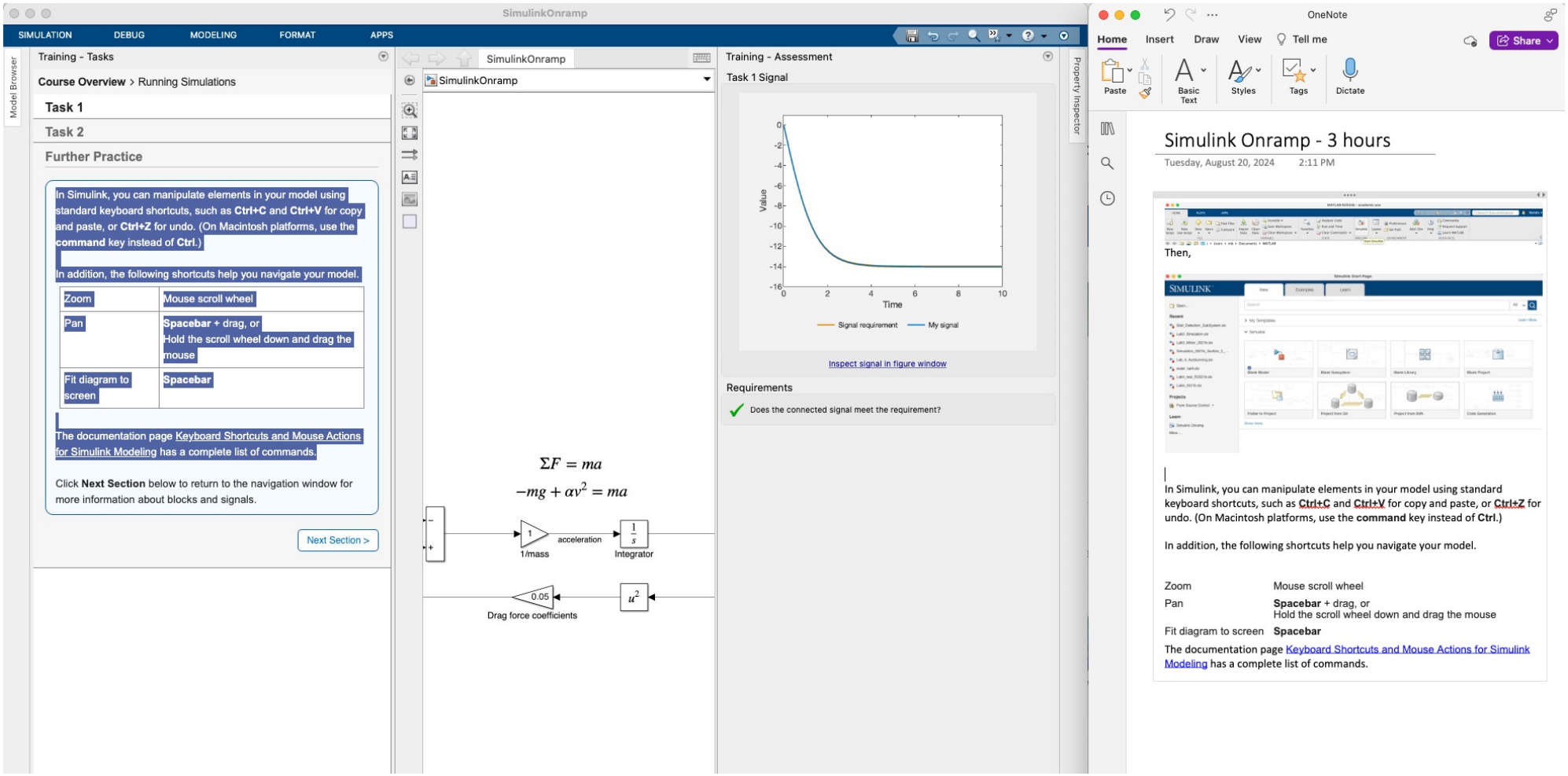
How to access Simulink?



How to access the training?



Make notes during these online MATLAB Training!

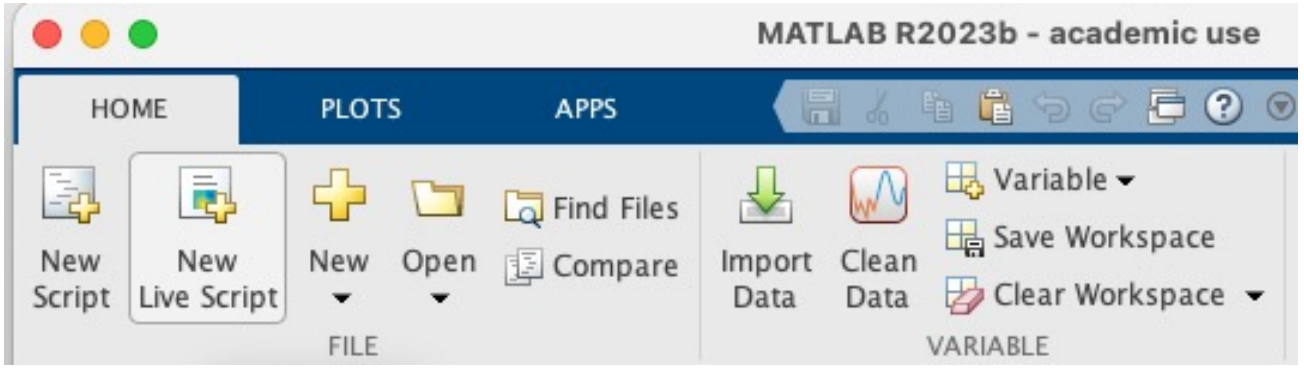


Report Guideline

- No formal report format required: you don't need to list equipment or explain procedures.
- Focus on what you observed and the insights you gained.
- Address all questions in the lab handout.
- Demonstrate your understanding of the lab material.
- Reports must be typed and submitted as a single PDF (separate images as solutions are not accepted)
  - Pages must be numbered
  - Figures and tables must have a caption
    - Table captions must be above tables, and figure captions below the figures
  - Equations must be numbered
- Upload all code files that you created to D2L, appropriately named
  - Simulink files (.slx format)
  - MATLAB files (.mlx format)
- Submit to D2L (do not compress files)
- No submission will be accepted by email

MATLAB Live scripts (`.mlx`)

- To include your code, output, and formatted text
- To create one, go to the Home tab and click New Live Script
- You need to answer the programming questions in a lab manual by writing programs in a **single** live script using MATLAB Live Editor. You must upload this one file to the D2L Dropbox folder https://www.mathworks.com/help/matlab/matlab_prog/create-live-scripts.html



MATLAB Live scripts (`.mlx`) Example

plotRand.mlx

Plot Random Data

This script plots a vector of random data and draws a horizontal line on the plot at the mean.

```
n = 50;
r = rand(n,1);
plot(r)

m = mean(r);
hold on
plot([0,n],[m,m])
hold off
title('Mean of Random Uniform Data')
```



Text

Code

Output

Source: Mathworks

Similar to a Python's Jupyter Notebook.

Figures

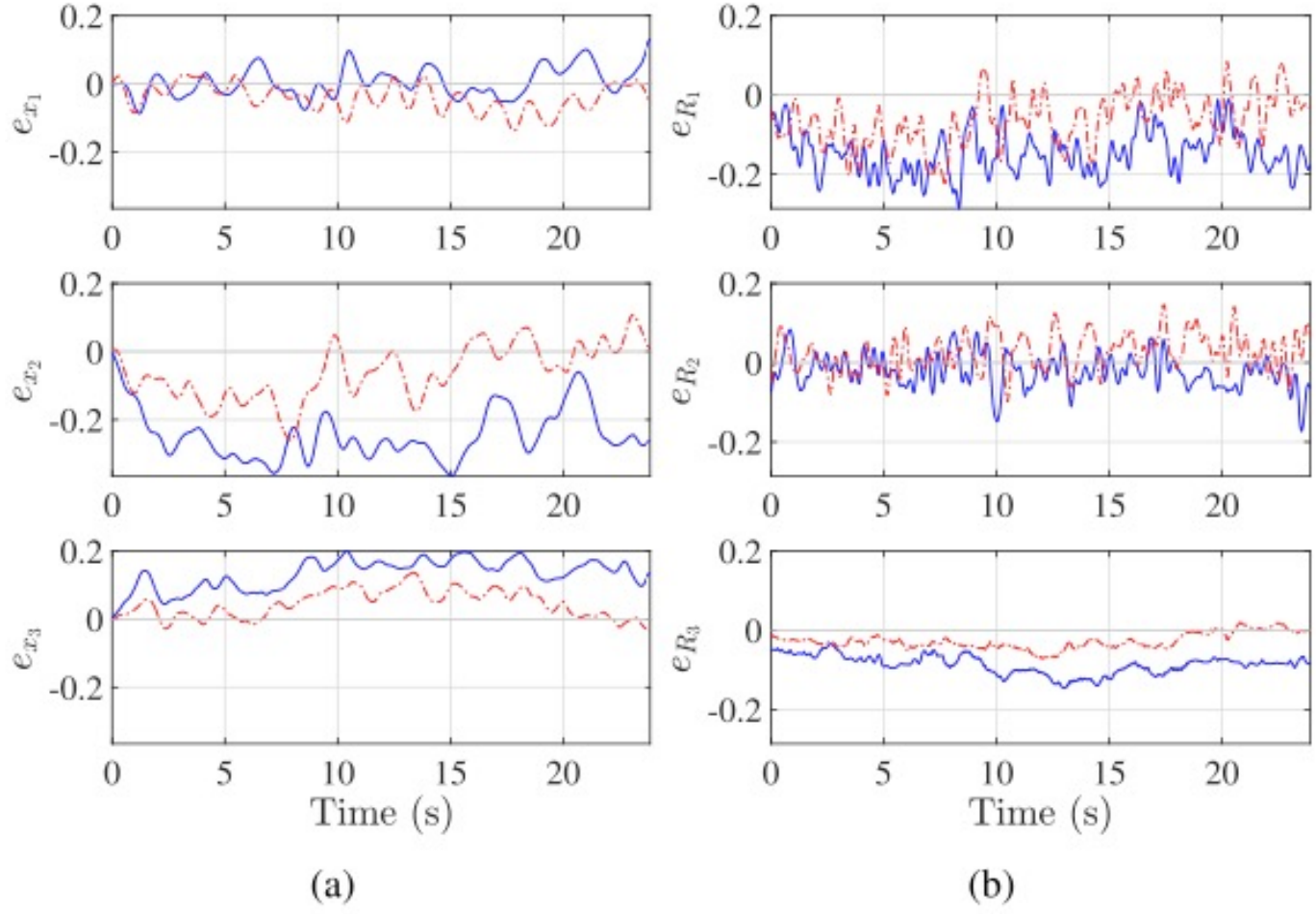
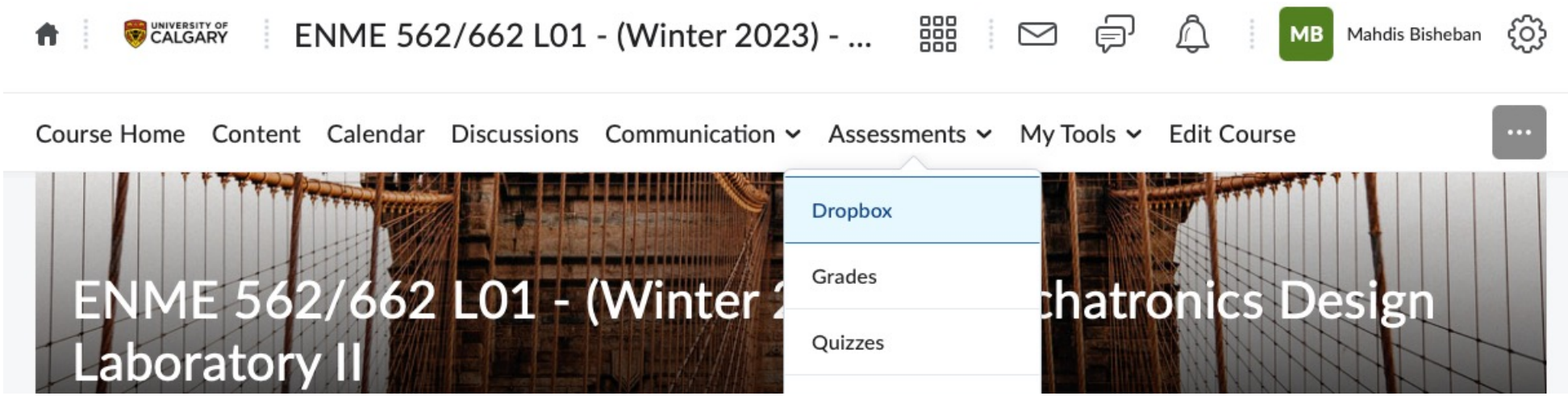


Fig. 13. Position tracking experiments: tracking errors (blue line: without disturbance rejection [12] and red dash-dotted line: adaptive controller). (a) Position error (m). (b) Attitude error.

References

- Include a numbered list of references, including laboratory handouts, lecture notes, reference books, journals, and webpages.
  - In-text citation is required.
  - Utilize quotation marks for any quoted material.
  - If a figure is taken from any source, it should be cited in the figure caption.
  - If an assumption is drawn from a book, webpage, etc., it must also be appropriately cited.

Report Submission



Ensure the group representative submits the final report.
One submission per group

Late Submission & Regarding

- Contact the instructor within 24 hours for missed lab reports, providing reasoning.
- If accepted, extra days may be granted with a 20% penalty per day.

- Re-grade requests must be submitted by email within ONE WEEK following the return of the graded material to the instructor and the TA
- A justification must accompany any re-grade request
- Simple re-grade requests are accepted at any time (e.g. adding errors, multiple deductions)

Communications

- All email communications must be with UCalgary email addresses
- Non-technical questions will be answered via email
- For technical questions, attend the office hours or ask TAs/Instructor during lab sessions



Team Contract

- **Agreement Required**
  All team members must agree on the team Contract.
- **Submit to D2L**
  Upload your team contact to the D2L
- **Accepted Formats**
  - Written Word document
  - Recorded video
  - Designed poster
  - Recorded audio
  - Other creative formats
- **Resources Available**
  - An example of a contract template is posted on D2L
  - This https://www.itpmetrics.com



Team Collaboration Guidelines for Lab 2-6

- Attend the designated lab section together.
- Engage actively in experiments, report preparation, and contribute ideas for analysis and discussions.
- Respond professionally and respectfully to all group correspondence within an agreed-upon timeframe.
- Ensure equal contribution from each team member.
- Complete the contribution table on the first page of the report.
- All group members must agree on the contribution percentage.

Team member	Experimental work and data collection	Analysis	MATLAB/Simulink code	Interpretation and validation	Report writing and editing	Overall contribution
Member 1:						_____%
Member 2:						_____%
Member 3:						_____%
Total						100%

How to Succeed in This Course?

- Attend Lectures
- Attend labs prepared
- Read
  - Textbook (Feedback Control of Dynamic Systems)
  - Lecture notes
  - Lab manuals
  - Lab presentations
  - Materials posted to “Readings” folder on D2L
  - Any other posted materials to D2L
- Review materials and lab manuals before attending lab sessions.
- Rotate roles within the team (e.g., writing MATLAB and Simulink code, answering questions, technical writing, etc.).
- Attend office hours if there are any questions
  - Office: MEB 512
  - If you have a class at the time of office hours, let me know. I’ll do my best to accommodate you

Schulich Makerspace Trainings

Schulich makerspaces offer many trainings. Book early; seats fill fast.

1 Electronics & PCB

- Soldering 1 — THT
- Soldering SMT I
- Soldering SMT II — Oven
- PCB

Build, populate and rework your own boards.

2 Rapid Prototyping

- Hands-on with 3D Printers
- Laser cutter

Enclosures, sensor mounts, brackets, front panels.

3 Machining & Structure

- Access training (required first)
- Lathe and Milling Machine
- CNC

Metal parts, shafts, bearing seats, precision fits.

4 Control & Code

- Robotics

Sensing, actuation and the software side of the loop.

https://schulich.libcal.com/calendar/workshops

Thank You!